Data Cleaning and Preparation

30

Common Data Issues in Surveys

- Misspellings
- Duplicate Responses
- Inconsistent Formats
- Outliers
- Missing Values

31

Why Does Data Go Missing?

- Data in surveys can go missing for a variety of reasons:
  - A participant skips a question.
  - A technical glitch prevents recording.
  - Sensitive questions (e.g., income, health) are left blank intentionally.

32

Types of Missingness

- **MCAR (Missing Completely At Random)**
  - The missingness is purely random and has **no relationship** to any data.
  - Example: A question is skipped due to internet disconnect.

33

Types of Missingness

- **MCAR (Missing Completely At Random)**
- **MAR (Missing At Random)**
  - The missingness is related to **other variables**, but not to the missing value itself.
  - Example: People who are younger are more likely to skip the income question.

34

Types of Missingness

- **MCAR (Missing Completely At Random)**
- **MAR (Missing At Random)**
- **MNAR (Missing Not At Random)**
  - The missingness is related to the missing value itself.
  - Example: People with **very low** income choose not to report it.

35

Techniques to Handle Missing Data

- Listwise Deletion
- Imputation
- Flag Missing Values

36

Exploratory Data Analysis (EDA) for Surveys

- EDA is the process of examining your dataset **visually and statistically** to understand its structure, patterns, anomalies, and potential relationships **before** building any models or drawing conclusions.
- EDA helps :
  - Understand who responded
  - See how people think
  - Spot trends across different demographics

37

Why EDA is Critical for Surveys?

- Surveys often have **categorical**, **ordinal**, and **textual** data.
- Survey data can be **biased, missing, or inconsistent**.
- EDA helps clean and prepare data for further analysis like clustering, regression, or dashboards.
- It allows stakeholders to **quickly get insights**.

38

Steps in EDA for Survey Data

- Understand Your Dataset
- Clean the Data
- Analyze One Variable at a Time
- Compare Two or More Variables
- Look for Patterns, Trends, and Outliers
- Visualize the Data
- Draw Preliminary Insights
- Prepare a Summary Report

39

Understand Your Dataset

- How many responses are there?
- What types of questions were asked?
- Check for missing or incomplete data

40

Clean the Data

- Fix inconsistent labels
- Remove duplicates
- Standardize formats

41

Analyze One Variable at a Time

- Categorical Variables (like Gender or Education)
- Rating Scales (1–5, etc.)
- Numerical Variables (like Age or Income)

42

Compare Two or More Variables

- Do younger students report higher satisfaction than older ones?
- Does income level affect satisfaction?
- Are there gender differences in study habits?
- Do certain education levels correlate with different survey responses?

43

Look for Patterns, Trends, and Outliers

- Are there **clear trends** by age, gender, or job role?
- Are there **extremely high or low values** that don't make sense?
- Are some groups **consistently more or less satisfied**?

44

Visualize the Data

- Convert raw numbers into understandable visuals:
  - **Pie charts** for proportion of categories (% of male vs female)
  - **Histograms** for seeing age or income distribution
  - **Box plots** to compare satisfaction across groups
  - **Word clouds** or keyword frequency for text-based responses

45

Draw Preliminary Insights

- Start making sense of the data:
  - What are the **key words**?
  - Are there any **red flags**?
  - What do these insights suggest for **next steps** or **decision-making**?

46

Prepare a Summary Report

- Summary of the dataset
- Key insights (with visuals or tables)
- Recommendations or next steps
- Limitations of the analysis (small sample, biased responses)

47

Case Study: Student Satisfaction Survey Online Learning Platform

- An xyz-tech company, launched a new online course on "Data Literacy for Beginners." After 6 weeks, they conducted a post-course survey to evaluate student experience.
- Imagine you have a dataset of **500 responses**.

48

Here's a sample of what 5 entries might look like:

Age	Gender	Country	Employment	Hours/Week	Satisfaction	Clarity	Recommend	Liked Most	Improve
21	Female	India	Student	5	4	4	Yes	Easy explanations	More real-world cases
35	Male	USA	Employed	3	3	2	No	Flexible timings	Instructor pacing
28	Female	Canada	Employed	4	5	5	Yes	Interactive examples	Add practice quizzes
19	Male	India	Student	8	2	3	No	Access anytime	Too much theory
42	Female	UK	Self-employed	2	4	4	Yes	Course structure	More peer discussion

49

EDA steps

- Step 1: Understand the data
- Step 2: Univariate Analysis
- Step 3: Bivariate Analysis

50